

4 Euclidean Approximation

In this chapter, we study approximation in *Euclidean* spaces. Therefore, $\mathcal{F}$ denotes a linear space, equipped with a *Euclidean norm* $\|\cdot\|$, i.e., $\|\cdot\|$ is defined by an inner product,

$$\|f\| = (f, f)^{1/2} \quad \text{for } f \in \mathcal{F}.$$

From the preceding chapter, we understand that Euclidean approximation has fundamental advantages, in particular for the existence and uniqueness of best approximations. We briefly summarize our previous results as follows.

Existence of best approximations: For a *Hilbert space* $\mathcal{F}$, i.e., $\mathcal{F}$ is *complete* with respect to $\|\cdot\|$, and a *closed* and *convex* subset $\mathcal{S} \subset \mathcal{F}$, there exists for any $f \in \mathcal{F}$ a best approximation $s^* \in \mathcal{S}$ to f .

Uniqueness best approximations: For *convex* $\mathcal{S} \subset \mathcal{F}$, a best approximation $s^* \in \mathcal{S}$ to $f \in \mathcal{F}$ is unique, due to the *strict convexity* of $\|\cdot\|$.

The above statements are based on Theorems 3.14, 3.28, and Corollary 3.39 from Chapter 3. Recall that for the existence and uniqueness of s^* , the *parallelogram identity* (3.1) plays a central role. Moreover, according to the Jordan-von Neumann theorem, Theorem 3.10, the parallelogram identity holds only in Euclidean spaces. Therefore, the problem of Euclidean approximation is fundamentally different from the problem of approximation in non-Euclidean spaces.

In this chapter, we explain further advantages of Euclidean approximation. To this end, we rely on the characterizations for best approximations. In particular, we make use of the Kolmogorov criterion, Corollary 3.55, in combination with the representation of Gâteaux derivatives for Euclidean norms from Theorem 3.62. This yields for *finite-dimensional* approximation spaces $\mathcal{S} \subset \mathcal{F}$ constructive methods to compute best approximations by *orthogonal projection* $\Pi : \mathcal{F} \rightarrow \mathcal{S}$ of $f \in \mathcal{F}$ on $\mathcal{S}$.

We treat two important special cases of Euclidean approximation: Firstly, the approximation of 2π -periodic continuous functions by trigonometric polynomials, where $\mathcal{F} = \mathcal{C}_{2\pi}$ and $\mathcal{S} = \mathcal{T}_n$. Secondly, the approximation of continuous functions by algebraic polynomials, in which case $\mathcal{F} = \mathcal{C}[a, b]$, for a compact interval $[a, b] \subset \mathbb{R}$, and $\mathcal{S} = \mathcal{P}_n$.

4.1 Construction of Best Approximations

In this section, we apply the characterizations for best approximations from the previous chapter to Euclidean spaces. To this end, we assume that the approximation space $\mathcal{S} \subset \mathcal{F}$ is a linear subspace of the Euclidean space $\mathcal{F}$. The application of the Kolmogorov criterion immediately provides the following fundamental result.

Theorem 4.1. *Let $\mathcal{F}$ be a Euclidean space with inner product $(\cdot, \cdot)$. Moreover, suppose $\mathcal{S} \subset \mathcal{F}$ is a convex subset of $\mathcal{F}$. Then the following statements are equivalent.*

- (a) $s^* \in \mathcal{S}$ is a best approximation to $f \in \mathcal{F} \setminus \mathcal{S}$.
- (b) We have $(s^* - f, s - s^*) \geq 0$ for all $s \in \mathcal{S}$.

Proof. Under the stated assumptions, the equivalence of the Kolmogorov criterion, Corollary 3.55, holds. Thereby, a best approximation $s^* \in \mathcal{S}$ to f is characterized by the necessary and sufficient condition

$$\|'_+(s^* - f, s - s^*) = \left(\frac{s^* - f}{\|s^* - f\|}, s - s^* \right) \geq 0 \quad \text{for all } s \in \mathcal{S},$$

using the representation of the Gâteaux derivative from Theorem 3.62. ■

Remark 4.2. If $\mathcal{S} \subset \mathcal{F}$ is a linear subspace of $\mathcal{F}$, then the variational inequality in statement (b) of Theorem 4.1 immediately leads us to the necessary and sufficient condition

$$(s^* - f, s) = 0 \quad \text{for all } s \in \mathcal{S}, \tag{4.1}$$

i.e., in this case, $s^* \in \mathcal{S}$ is a best approximation to $f \in \mathcal{F} \setminus \mathcal{S}$, if and only if the orthogonality relation $s^* - f \perp \mathcal{S}$ holds. □

Note that the equivalence statement in Remark 4.2 identifies a best approximation $s^* \in \mathcal{S}$ to $f \in \mathcal{F}$ as the unique *orthogonal projection* of f onto $\mathcal{S}$. In Section 4.2, we will study the projection operator $\Pi : \mathcal{F} \rightarrow \mathcal{S}$, which assigns every $f \in \mathcal{F}$ to its unique best approximation $s^* \in \mathcal{S}$ in more detail.

Before doing so, we first use the orthogonality in (4.1) to characterize best approximations $s^* \in \mathcal{S}$ for convex subsets $\mathcal{S} \subset \mathcal{F}$. To this end, we work with the dual characterization of Theorem 3.46.

Theorem 4.3. *Let $\mathcal{F}$ be a Euclidean space with inner product $(\cdot, \cdot)$ and let $\mathcal{S} \subset \mathcal{F}$ be a convex subset of $\mathcal{F}$. Moreover, suppose that $s^* \in \mathcal{S}$ satisfies $s^* - f \perp \mathcal{S}$. Then, s^* is the unique best approximation to f .*

Proof. The linear functional $\varphi \in \mathcal{F}'$, defined as

$$\varphi(u) = \left(\frac{s^* - f}{\|s^* - f\|}, u \right) \quad \text{for } u \in \mathcal{F},$$

satisfies all three conditions from the dual characterization of Theorem 3.46: Indeed, the first condition, $\|\varphi\| = 1$, follows from the Cauchy¹-Schwarz² inequality,

$$|\varphi(u)| = \left| \left(\frac{s^* - f}{\|s^* - f\|}, u \right) \right| \leq \frac{\|s^* - f\|}{\|s^* - f\|} \cdot \|u\| = \|u\| \quad \text{for all } u \in \mathcal{F},$$

where equality holds for $u = s^* - f \in \mathcal{F}$, since

$$\varphi(s^* - f) = \left(\frac{s^* - f}{\|s^* - f\|}, s^* - f \right) = \frac{\|s^* - f\|^2}{\|s^* - f\|} = \|s^* - f\|.$$

Therefore, φ also satisfies the second condition in Theorem 3.46. By $s^* - f \perp \mathcal{S}$ we have

$$\varphi(s) = \left(\frac{s^* - f}{\|s^* - f\|}, s \right) = 0 \quad \text{for all } s \in \mathcal{S},$$

and so φ finally satisfies the third condition in Theorem 3.46.

In conclusion, s^* is a best approximation to f . The uniqueness of s^* follows from the strict convexity of the Euclidean norm $\|\cdot\| = (\cdot, \cdot)^{1/2}$. ■

Now we consider the special case of Euclidean approximation by *finite-dimensional* approximation spaces $\mathcal{S}$. Therefore, suppose that $\mathcal{S} \subset \mathcal{F}$ is a linear subspace with $\dim(\mathcal{S}) < \infty$. According to Corollary 3.8, there exists for any $f \in \mathcal{F}$ a best approximation $s^* \in \mathcal{S}$ to f and, moreover, s^* is unique, due to Theorem 3.37.

Suppose that $\mathcal{S}$ is spanned by $n \in \mathbb{N}$ basis elements $\{s_1, \dots, s_n\} \subset \mathcal{F}$, i.e.,

$$\mathcal{S} = \text{span}\{s_1, \dots, s_n\} \subset \mathcal{F}$$

so that $\dim(\mathcal{S}) = n < \infty$. To compute the unique best approximation $s^* \in \mathcal{S}$ to some $f \in \mathcal{F}$ we utilize the representation

$$s^* = \sum_{j=1}^n c_j^* s_j \in \mathcal{S}. \quad (4.2)$$

Now the (necessary and sufficient) orthogonality condition $s^* - f \perp \mathcal{S}$ from Remark 4.2, or, in (4.1) is equivalent to the requirement

$$(s^*, s_k) = (f, s_k) \quad \text{for all } 1 \leq k \leq n.$$

Therefore, the representation for s^* in (4.2) leads us to the n linear conditions

$$\sum_{j=1}^n c_j^* (s_j, s_k) = (f, s_k) \quad \text{for } 1 \leq k \leq n$$

¹ AUGUSTIN-LOUIS CAUCHY (1789-1857), French mathematician

² HERMANN AMANDUS SCHWARZ (1843-1921), German mathematician

and so to the linear equation system

$$\begin{bmatrix} (s_1, s_1) & (s_2, s_1) & \cdots & (s_n, s_1) \\ (s_1, s_2) & (s_2, s_2) & \cdots & (s_n, s_2) \\ \vdots & \vdots & \ddots & \vdots \\ (s_1, s_n) & (s_2, s_n) & \cdots & (s_n, s_n) \end{bmatrix} \cdot \begin{bmatrix} c_1^* \\ c_2^* \\ \vdots \\ c_n^* \end{bmatrix} = \begin{bmatrix} (f, s_1) \\ (f, s_2) \\ \vdots \\ (f, s_n) \end{bmatrix},$$

or, in short,

$$Gc^* = b \quad (4.3)$$

with the **Gram**³ matrix $G = ((s_j, s_k))_{1 \leq j, k \leq n} \in \mathbb{R}^{n \times n}$, the unknown coefficient vector $c^* = (c_1^*, \dots, c_n^*)^T \in \mathbb{R}^n$ of s^* in (4.2) and the right hand side $b = ((f, s_1), \dots, (f, s_n))^T \in \mathbb{R}^n$. Therefore, the solution $c^* \in \mathbb{R}^n$ of the linear system (4.3) yields the unknown coefficients of s^* in (4.2).

Due to the existence and uniqueness of the best approximation s^* , the Gram matrix G must be regular. We specialize this statement on G as follows.

Theorem 4.4. *The Gram matrix G in (4.3) is symmetric positive definite.*

Proof. The symmetry of G follows from the symmetry of the inner product, whereby $(s_j, s_k) = (s_k, s_j)$ for all $1 \leq j, k \leq n$. Moreover, G is positive definite, which also immediately follows from the properties of the inner product,

$$c^T G c = \sum_{j,k=1}^n c_j c_k (s_j, s_k) = \left(\sum_{j=1}^n c_j s_j, \sum_{k=1}^n c_k s_k \right) = \left\| \sum_{j=1}^n c_j s_j \right\|^2 > 0$$

for all $c = (c_1, \dots, c_n)^T \in \mathbb{R}^n \setminus \{0\}$. ■

Given our investigations in this section, the problem of Euclidean approximation by finite-dimensional approximation spaces $\mathcal{S}$ seems to be solved. The unique best approximation s^* can be determined by the unique solution of the linear system (4.3), i.e., computing s^* is equivalent to solving (4.3).

But note that we have not posed any conditions on the basis of $\mathcal{S}$, yet.

Next, we show that by suitable choices for bases of $\mathcal{S}$, we can avoid the linear system (4.3). Indeed, for an **orthogonal basis** $\{s_1, \dots, s_n\}$ of $\mathcal{S}$, i.e.,

$$(s_j, s_k) = \begin{cases} 0 & \text{for } j \neq k, \\ \|s_j\|^2 > 0 & \text{for } j = k, \end{cases}$$

the Gram matrix G is a diagonal matrix,

$$G = \text{diag}(\|s_1\|^2, \dots, \|s_n\|^2) = \begin{bmatrix} \|s_1\|^2 & & & \\ & \|s_2\|^2 & & \\ & & \ddots & \\ & & & \|s_n\|^2 \end{bmatrix},$$

³ JØRGEN PEDERSEN GRAM (1850-1916), Danish mathematician

in which case the solution c^* of (4.3) is given by

$$c^* = \left(\frac{(f, s_1)}{\|s_1\|^2}, \dots, \frac{(f, s_n)}{\|s_n\|^2} \right)^T \in \mathbb{R}^n.$$

For an **orthonormal basis** $\{s_1, \dots, s_n\}$ of $\mathcal{S}$, i.e., $(s_j, s_k) = \delta_{jk}$, the Gram matrix G is the identity matrix, $G = I_n \in \mathbb{R}^{n \times n}$, in which case

$$c^* = ((f, s_1), \dots, (f, s_n))^T \in \mathbb{R}^n.$$

In the following of this chapter, we develop suitable constructions and characterizations for orthogonal bases in relevant applications. Before doing so, we summarize the discussion of this section, and, moreover, we derive a few elementary properties of orthogonal bases.

4.2 Orthogonal Bases and Orthogonal Projections

From our discussion in the previous section, we can explicitly represent, for a fixed orthogonal basis (orthonormal basis), $\{s_1, \dots, s_n\}$ of $\mathcal{S}$, the unique best approximation $s^* \in \mathcal{S}$ to f , for any $f \in \mathcal{F}$.

Theorem 4.5. *Let $\mathcal{F}$ be a Euclidean space with inner product $(\cdot, \cdot)$. Moreover, let $\mathcal{S} \subset \mathcal{F}$ be a finite-dimensional linear subspace with orthogonal basis $\{s_1, \dots, s_n\}$. Then, for any $f \in \mathcal{F}$,*

$$s^* = \sum_{j=1}^n \frac{(f, s_j)}{\|s_j\|^2} s_j \in \mathcal{S} \quad (4.4)$$

is the unique best approximation to f . For the special case of an orthonormal basis $\{s_1, \dots, s_n\}$ for $\mathcal{S}$ we have the representation

$$s^* = \sum_{j=1}^n (f, s_j) s_j \in \mathcal{S}.$$

■

Now we study the linear and surjective operator $\Pi : \mathcal{F} \rightarrow \mathcal{S}$, which maps any element $f \in \mathcal{F}$ to its unique best approximation $s^* \in \mathcal{S}$. But first we note that the optimality of the best approximation $s^* = \Pi(f)$ immediately implies the *stability estimate*

$$\|(I - \Pi)(f)\| \leq \|f - s\| \quad \text{for all } f \in \mathcal{F}, s \in \mathcal{S} \quad (4.5)$$

where I denotes the identity on $\mathcal{F}$. Moreover, for $f = s$ in (4.5) we get

$$\Pi(s) = s \quad \text{for all } s \in \mathcal{S}$$

and so Π is a **projection operator**, i.e., $\Pi \circ \Pi = \Pi$. By the characterization of the best approximation $s^* = \Pi(f) \in \mathcal{S}$ in (4.1), the operator Π is an **orthogonal projection**, since

$$f - \Pi(f) = (I - \Pi)(f) \perp \mathcal{S} \quad \text{for all } f \in \mathcal{F},$$

i.e., the linear operator $I - \Pi : \mathcal{F} \rightarrow \mathcal{S}^\perp$ maps onto the *orthogonal complement* $\mathcal{S}^\perp \subset \mathcal{F}$ of $\mathcal{S}$ in $\mathcal{F}$. Moreover, $I - \Pi$ is also a projection operator, since for any $f \in \mathcal{F}$, we have

$$\begin{aligned} ((I - \Pi) \circ (I - \Pi))(f) &= (I - \Pi)(f - \Pi(f)) \\ &= f - \Pi(f) - \Pi(f) + (\Pi \circ \Pi)(f) \\ &= f - \Pi(f) = (I - \Pi)(f). \end{aligned}$$

The orthogonality of Π immediately implies another well-known result.

Theorem 4.6. *The Pythagoras⁴ theorem*

$$\|f - \Pi(f)\|^2 + \|\Pi(f)\|^2 = \|f\|^2 \quad \text{for all } f \in \mathcal{F} \quad (4.6)$$

holds.

Proof. For $f \in \mathcal{F}$ we have

$$\begin{aligned} \|f\|^2 &= \|f - \Pi(f) + \Pi(f)\|^2 \\ &= \|f - \Pi(f)\|^2 + 2(f - \Pi(f), \Pi(f)) + \|\Pi(f)\|^2 \\ &= \|f - \Pi(f)\|^2 + \|\Pi(f)\|^2. \end{aligned}$$

■

The Pythagoras theorem implies two further *stability results*.

Corollary 4.7. *For $I \neq \Pi$ the stability estimates*

$$\|(I - \Pi)(f)\| \leq \|f\| \quad \text{and} \quad \|\Pi(f)\| \leq \|f\| \quad \text{for all } f \in \mathcal{F} \quad (4.7)$$

hold. In particular, we have

$$\|I - \Pi\| = 1 \quad \text{and} \quad \|\Pi\| = 1$$

for the operator norms of $I - \Pi$ and Π .

Proof. The stability estimates in (4.7) follow directly from the Pythagoras theorem, Theorem 4.6. In the first inequality in (4.7), we have equality for every element $f - \Pi(f) \in \mathcal{S}^\perp$, whereas in the second inequality, we have equality for every $s \in \mathcal{S}$. Thereby, the operator norms of $I - \Pi$ and Π are already determined by

$$\|I - \Pi\| = \sup_{f \neq 0} \frac{\|(I - \Pi)(f)\|}{\|f\|} = 1 \quad \text{and} \quad \|\Pi\| = \sup_{f \neq 0} \frac{\|\Pi(f)\|}{\|f\|} = 1.$$

■

⁴ PYTHAGORAS OF SAMOS (around 570-510 BC), ancient Greek philosopher

Next, we compute for $f \in \mathcal{F}$ the norm $\| \Pi(f) \|$ of $\Pi(f) = s^*$. To this end, we utilize, for a fixed orthogonal basis $\{s_1, \dots, s_n\}$ of $\mathcal{S}$ the representation in (4.4), whereby

$$\Pi(f) = \sum_{j=1}^n \frac{(f, s_j)}{\|s_j\|^2} s_j \in \mathcal{S} \quad \text{for } f \in \mathcal{F}. \quad (4.8)$$

In particular, for $s \in \mathcal{S}$, we have the representation

$$\Pi(s) = s = \sum_{j=1}^n \frac{(s, s_j)}{\|s_j\|^2} s_j \in \mathcal{S} \quad \text{for all } s \in \mathcal{S}. \quad (4.9)$$

Theorem 4.8. *Let $\{s_1, \dots, s_n\} \subset \mathcal{S}$ be an orthogonal basis of $\mathcal{S}$. Then, the Parseval⁵ identity*

$$(\Pi(f), \Pi(g)) = \sum_{j=1}^n \frac{(f, s_j)(g, s_j)}{\|s_j\|^2} \quad \text{for all } f, g \in \mathcal{F} \quad (4.10)$$

holds, where in particular

$$\|\Pi(f)\|^2 = \sum_{j=1}^n \frac{|(f, s_j)|^2}{\|s_j\|^2} \quad \text{for all } f \in \mathcal{F}. \quad (4.11)$$

Proof. By the representation of Π in (4.8), we have

$$\begin{aligned} (\Pi(f), \Pi(g)) &= \left(\sum_{j=1}^n \frac{(f, s_j)}{\|s_j\|^2} s_j, \sum_{k=1}^n \frac{(g, s_k)}{\|s_k\|^2} s_k \right) \\ &= \sum_{j,k=1}^n \frac{(f, s_j)}{\|s_j\|^2} \frac{(g, s_k)}{\|s_k\|^2} (s_j, s_k) = \sum_{j=1}^n \frac{(f, s_j)(g, s_j)}{\|s_j\|^2} \end{aligned}$$

for all $f, g \in \mathcal{F}$. For $f = g$ we obtain the stated representation in (4.11). ■

We finally add another important result.

Theorem 4.9. *Let $\{s_1, \dots, s_n\} \subset \mathcal{S}$ be an orthogonal basis of $\mathcal{S}$. Then the Bessel⁶ inequality*

$$\|\Pi(f)\|^2 = \sum_{j=1}^n \frac{|(f, s_j)|^2}{\|s_j\|^2} \leq \|f\|^2 \quad \text{for all } f \in \mathcal{F} \quad (4.12)$$

holds. Moreover, we have the identity

$$\|f - \Pi(f)\|^2 = \|f\|^2 - \sum_{j=1}^n \frac{|(f, s_j)|^2}{\|s_j\|^2} \leq \|f\|^2 \quad \text{for all } f \in \mathcal{F}.$$

⁵ MARC-ANTOINE PARSEVAL DES CHÊNES (1755-1836), French mathematician

⁶ FRIEDRICH WILHELM BESSEL (1784-1846), German astronomer, mathematician

Proof. The Bessel inequality follows from the second stability estimate in (4.7) in combination with the representation in (4.11). The second statement follows from the Pythagoras theorem (4.6) and the representation (4.11). ■

4.3 Fourier Partial Sums

In this section, we study one concrete example for Euclidean approximation. In this particular case, we wish to approximate a continuous 2π -periodic function by real-valued trigonometric polynomials. To this end, we equip the linear space of all *real-valued* continuous 2π -periodic functions

$$\mathcal{C}_{2\pi}^{\mathbb{R}} \equiv \mathcal{C}_{2\pi} = \{f : \mathbb{R} \longrightarrow \mathbb{R} \mid f \in \mathcal{C}(\mathbb{R}) \text{ and } f(x) = f(x + 2\pi) \text{ for all } x \in \mathbb{R}\}$$

with the inner product

$$(f, g)_{\mathbb{R}} = \frac{1}{\pi} \int_0^{2\pi} f(x)g(x) \, dx \quad \text{for } f, g \in \mathcal{C}_{2\pi}, \quad (4.13)$$

which by $\|\cdot\|_{\mathbb{R}} = (\cdot, \cdot)_{\mathbb{R}}^{1/2}$ defines a Euclidean norm on $\mathcal{C}_{2\pi}$, so that

$$\|f\|_{\mathbb{R}}^2 = \frac{1}{\pi} \int_0^{2\pi} |f(x)|^2 \, dx \quad \text{for } f \in \mathcal{C}_{2\pi}.$$

Therefore, $\mathcal{C}_{2\pi}$ with $\|\cdot\|_{\mathbb{R}} = (\cdot, \cdot)_{\mathbb{R}}^{1/2}$ is a Euclidean space.

For the approximation space, we consider choosing the linear space of all real-valued trigonometric polynomials of degree at most $n \in \mathbb{N}_0$,

$$\mathcal{T}_n \equiv \mathcal{T}_n^{\mathbb{R}} = \text{span} \left\{ \frac{1}{\sqrt{2}}, \cos(j \cdot), \sin(j \cdot) \mid 1 \leq j \leq n \right\} \quad \text{for } n \in \mathbb{N}_0.$$

Therefore, by using the notations introduced at the outset of this chapter, we consider the special case of the Euclidean space $\mathcal{F} = \mathcal{C}_{2\pi}$, equipped with the norm $\|\cdot\|_{\mathbb{R}} = (\cdot, \cdot)_{\mathbb{R}}^{1/2}$, and the linear approximation space $\mathcal{S} = \mathcal{T}_n \subset \mathcal{C}_{2\pi}$ of finite dimension $\dim(\mathcal{T}_n) = 2n + 1$, for $n \in \mathbb{N}_0$.

Remark 4.10. In the following chapters, we also use *complex-valued* trigonometric polynomials from $\mathcal{T}_n^{\mathbb{C}}$ for the approximation of *complex-valued* continuous 2π -periodic functions from

$$\mathcal{C}_{2\pi}^{\mathbb{C}} = \{f : \mathbb{R} \longrightarrow \mathbb{C} \mid f \in \mathcal{C}(\mathbb{R}) \text{ and } f(x) = f(x + 2\pi) \text{ for all } x \in \mathbb{R}\}.$$

In that case, we equip $\mathcal{C}_{2\pi}^{\mathbb{C}}$ with the inner product

$$(f, g)_{\mathbb{C}} = \frac{1}{2\pi} \int_0^{2\pi} f(x)\overline{g(x)} \, dx \quad \text{for } f, g \in \mathcal{C}_{2\pi}^{\mathbb{C}}, \quad (4.14)$$

and thereby obtain the Euclidean norm $\|\cdot\|_{\mathbb{C}} = (\cdot, \cdot)_{\mathbb{C}}^{1/2}$ on $\mathcal{C}_{2\pi}^{\mathbb{C}}$. The different scalar factors, $1/\pi$ for $(\cdot, \cdot)_{\mathbb{R}}$ in (4.13) and $1/(2\pi)$ for $(\cdot, \cdot)_{\mathbb{C}}$ in (4.14), will be useful later. To keep notations simple, we will from now use $(\cdot, \cdot) = (\cdot, \cdot)_{\mathbb{R}}$ and $\|\cdot\| = \|\cdot\|_{\mathbb{R}}$ for the inner product (4.13) and the norm on $\mathcal{C}_{2\pi} \equiv \mathcal{C}_{2\pi}^{\mathbb{R}}$. □